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Q.1 Show that the grammar is ambiguous

Given Grammar

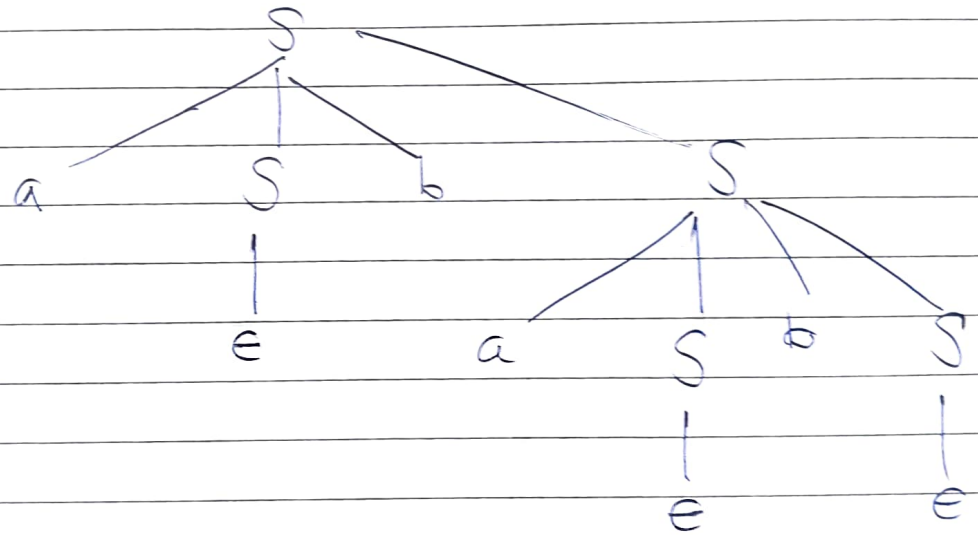
$S \rightarrow aSbS / bSaS / \epsilon$

A CFG is ambiguous if at least one string in its language has two different parse trees.

Consider: $w = abab$

$S \rightarrow aSbS \rightarrow abS \rightarrow abaSbS \rightarrow ababS \rightarrow abab$

Parse tree 1 for abab



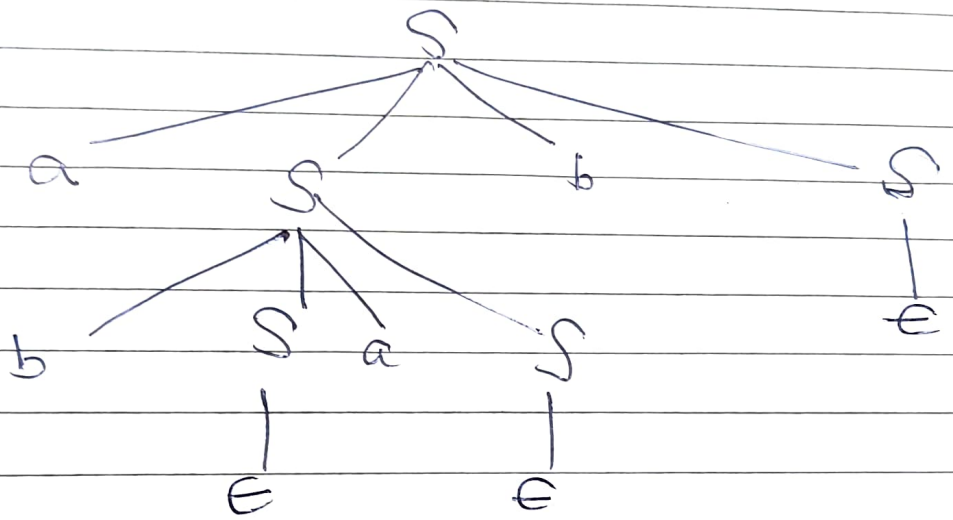
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Second LMD

$S \rightarrow aSbS \rightarrow abSaS \; SbS \rightarrow abaSbS \rightarrow ababS$
$\rightarrow abab$

2nd Parse tree



Since the same string abab has two different parse trees, the grammar is ambiguous

2 (a) Convert grammar into CNF.

$S \rightarrow \sim S[S \supset S] / P / q$

In CNF, every production must be of the form $A \rightarrow BC$ or $A \rightarrow a$

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S1: Replace terminals appearing in longer Productions

$N \rightarrow \sim$, $L \rightarrow [$, $I \rightarrow \supset$, $R \rightarrow]$

Then NS becomes $\sim S$ and $[S \supset S]$ becomes $LSISR$.

S2: Break the long right-hand side into pairs

$X_1 \rightarrow SX_2$

$X_2 \rightarrow IX_3$

$X_3 \rightarrow SR$

$\therefore S \rightarrow LSISR$ can be represented by $S \rightarrow LX_1$

Final CNF

$S \rightarrow NS / LX_1 / p / q$

$X_1 \rightarrow SX_2$

$X_2 \rightarrow IX_3$

$X_3 \rightarrow SR$

$N \rightarrow \sim$

$L \rightarrow [$

$I \rightarrow \supset$

$R \rightarrow]$

All productions are now of form $A \rightarrow BC$ or $A \rightarrow a$

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Q2. (b)

Convert in CNF

$S \rightarrow ASB$
$A \rightarrow aAS / a / \epsilon$
$B \rightarrow SbS / A / bb$

S1: And nullable variables and remove ϵ production.

$S \rightarrow ASB / SB / AS$
$A \rightarrow aAS / aS / a$
$B \rightarrow SbS / A / bb$

S2: Romive unit production

~~$A \rightarrow B$~~ $B \rightarrow A$

$B \rightarrow SbS / bb / aAS / aS / a$

S3: $S \rightarrow ASB / SB / AS$
$A \rightarrow CAS / CS / a$
$B \rightarrow SDS / DD / CAS / CS / a$
$C \rightarrow a$
$D \rightarrow b$

final CNF

$S \rightarrow AX_1 / SB / AS$
~~$X_1 \rightarrow SB$~~
$A \rightarrow CX_2 / CS / a$
$X_2 \rightarrow AS$
$B \rightarrow SX_3 / DD / CX_2 / CS / a$
$X_3 \rightarrow DS$
$C \rightarrow a$, $D \rightarrow b$

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